Optimization TA 7

Kacper Krasowski

Exercise 7.5

Problem

Any hyperplane in $\mathbb{R}^n$ corresponds to the level set of a linear functional $f : X \to \mathbb{R}$ with $f(\mathbf{x}) = \mathbf{p} \cdot \mathbf{x}$ and $\mathbf{p}$ nonzero:

$$H(\mathbf{p}, c) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p} \cdot \mathbf{x} = c\}.$$

Show that $\mathbf{p}$ is orthogonal to any vector $(\mathbf{x} - \mathbf{y})$ with $\mathbf{x}, \mathbf{y} \in H(\mathbf{p}, c)$. Describe the set of hyperplanes which support $\mathbb{R}^2_+$ at $\mathbf{0}$ (see section 3.9 in Carter for supporting hyperplanes).

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$$\boxed{\mathbf{p} \cdot (\mathbf{x} - \mathbf{y}) = 0 \text{ for every } \mathbf{x}, \mathbf{y} \in H(\mathbf{p}, c)}$$

$$\boxed{\text{Find every } H(\mathbf{p}, c) \text{ that supports } \mathbb{R}^2_+ \text{ at } \mathbf{0}}$$

Exercise 7.5: Definitions

Definitions

Orthogonal: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\mathbf{x} \cdot \mathbf{y} = 0$, where $\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + \cdots + x_n y_n$.

Halfspaces: The hyperplane $H(\mathbf{p}, c)$ divides $\mathbb{R}^n$ into

$$H^+(\mathbf{p}, c) = \{\mathbf{x} : \mathbf{p} \cdot \mathbf{x} \geq c\}, \qquad H^-(\mathbf{p}, c) = \{\mathbf{x} : \mathbf{p} \cdot \mathbf{x} \leq c\}.$$

Supporting hyperplane: $H(\mathbf{p}, c)$ supports a set S at $\mathbf{x}_0 \in S$ if

- $\mathbf{x}_0 \in H(\mathbf{p}, c)$;
- S lies on one side of $H(\mathbf{p}, c)$: $S \subseteq H^+(\mathbf{p}, c)$ or $S \subseteq H^-(\mathbf{p}, c)$.

Exercise 7.5: Sketch of the Proof — First Question

$$\boxed{\mathbf{p} \cdot (\mathbf{x} - \mathbf{y}) = 0 \text{ for every } \mathbf{x}, \mathbf{y} \in H(\mathbf{p}, c)}$$

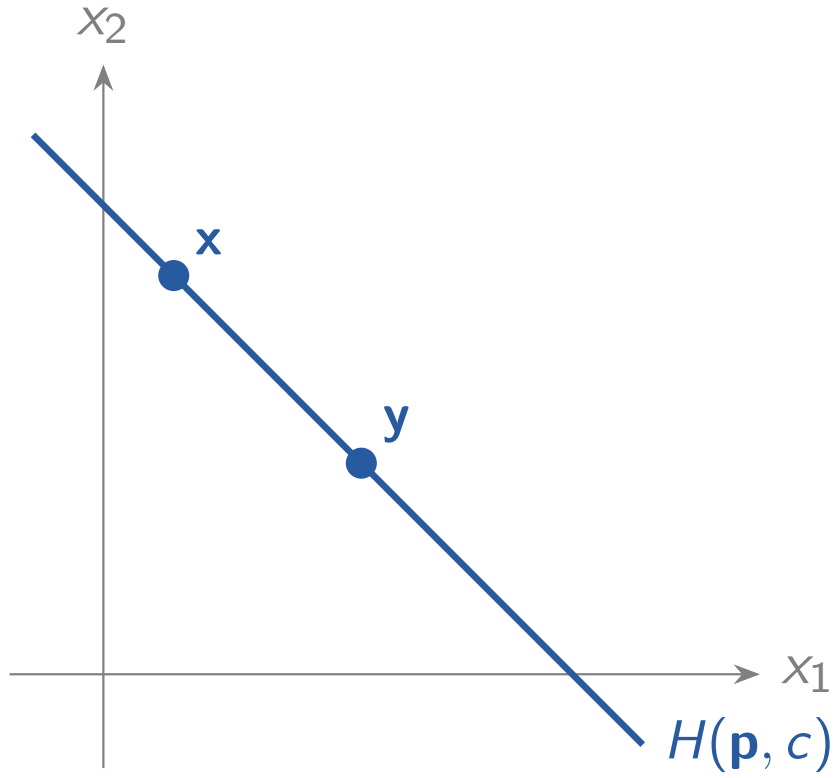
1. Expand $\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})$ using the linearity of the inner product.

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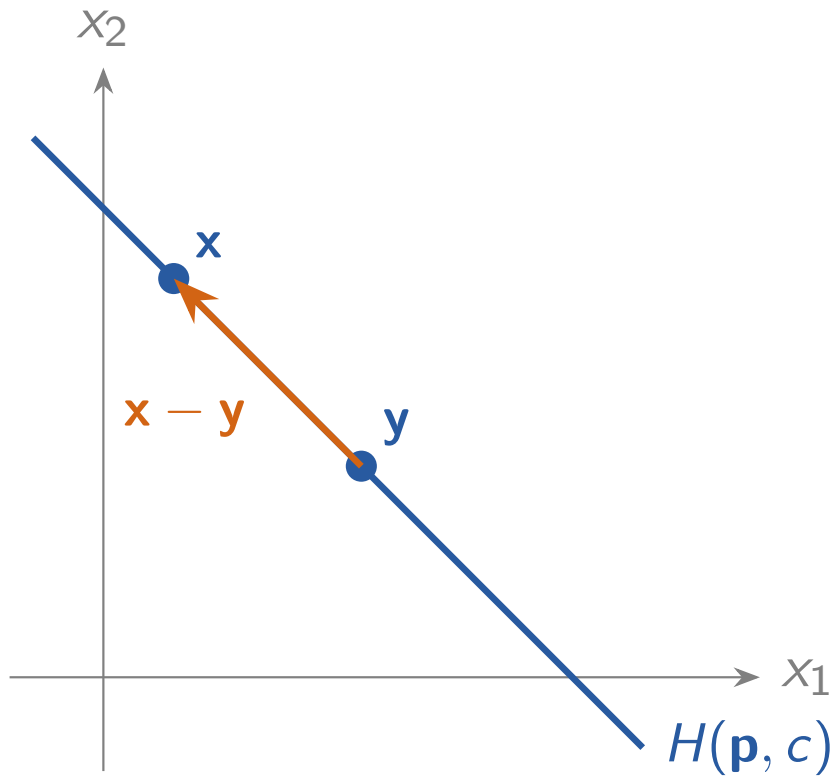
1. Expand $\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})$ using the linearity of the inner product.
2. Use that $\mathbf{x}$ and $\mathbf{y}$ lie in $H(\mathbf{p}, c)$.

Exercise 7.5: Intuition



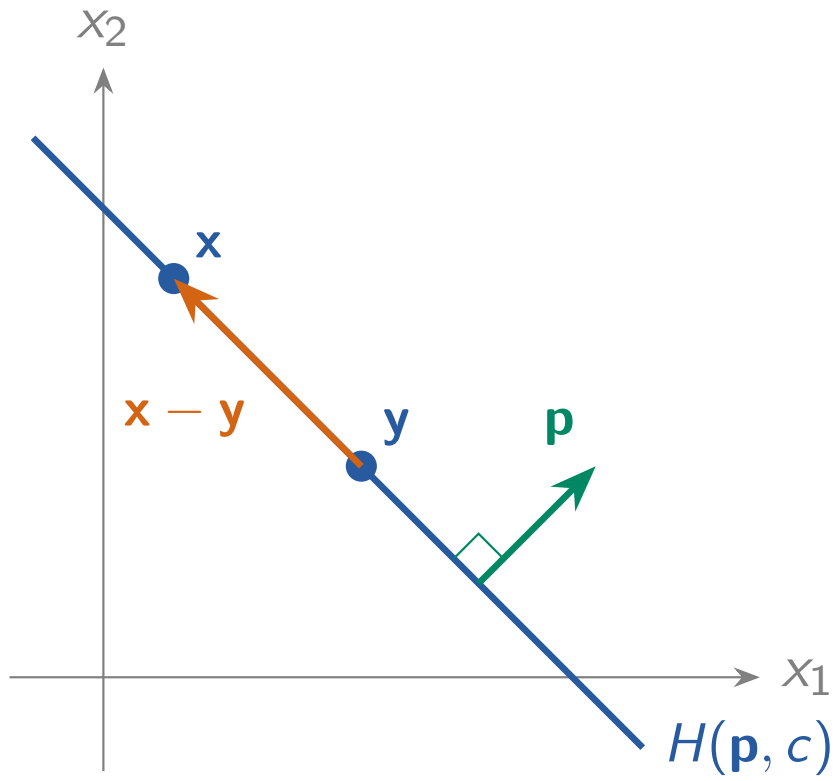
Two points $\mathbf{x}, \mathbf{y}$ on a hyperplane $H(\mathbf{p}, c)$ in $\mathbb{R}^2$.

Exercise 7.5: Intuition



The difference $\mathbf{x} - \mathbf{y}$ points along the hyperplane.

Exercise 7.5: Intuition



p is the normal: it is perpendicular to the hyperplane.

Exercise 7.5: Sketch of the Proof — Second Question

Find every $H(\mathbf{p}, c)$ that supports $\mathbb{R}^2_+$ at $\mathbf{0}$

1. Use $\mathbf{0} \in H(\mathbf{p}, c)$ to determine c.

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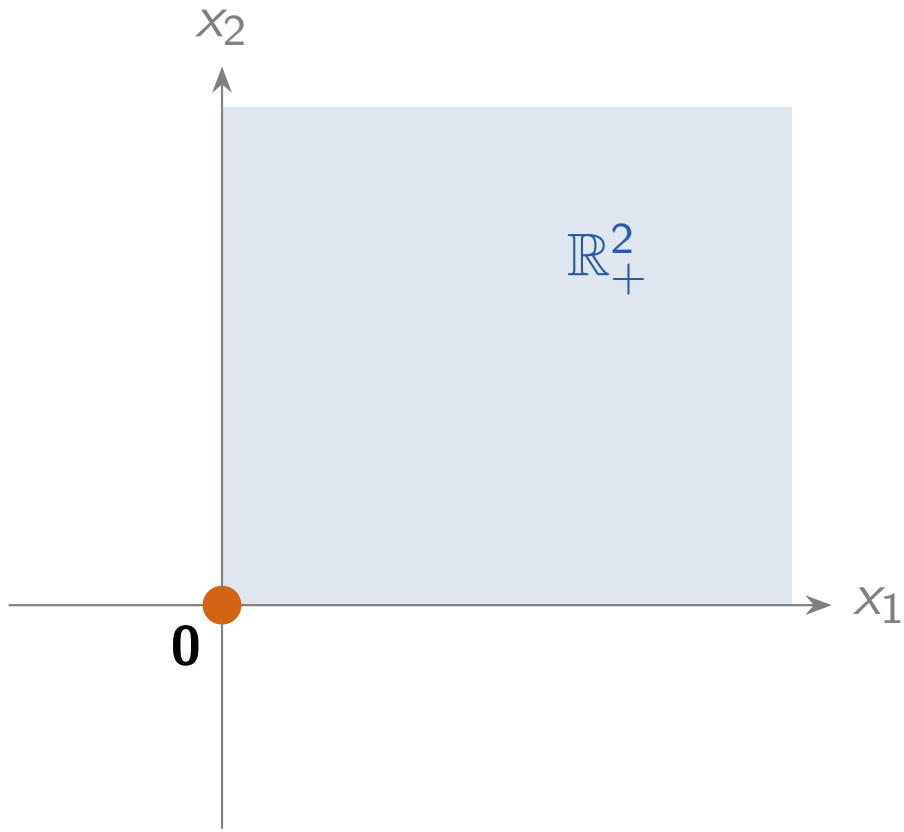
1. Use $\mathbf{0} \in H(\mathbf{p}, c)$ to determine c.
2. Find the conditions on $\mathbf{p}$ under which $\mathbb{R}^2_+$ lies in one of the halfspaces.

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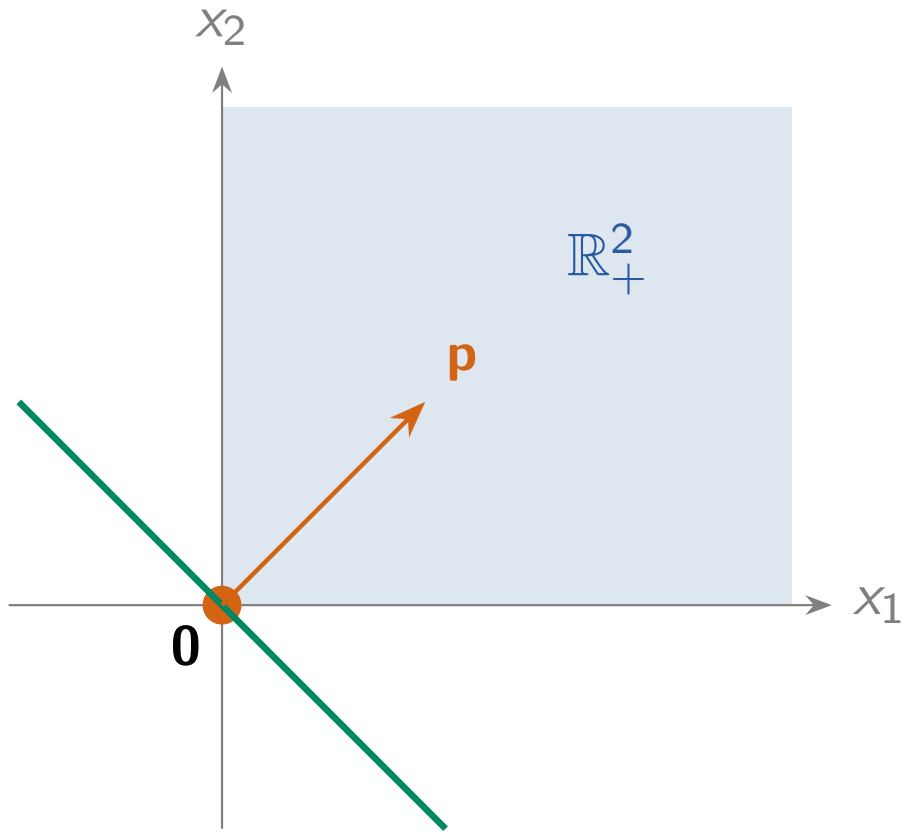
1. Use $\mathbf{0} \in H(\mathbf{p}, c)$ to determine c.
2. Find the conditions on $\mathbf{p}$ under which $\mathbb{R}^2_+$ lies in one of the halfspaces.
3. Describe the resulting hyperplanes.

Exercise 7.5: Intuition — Supports at the Origin



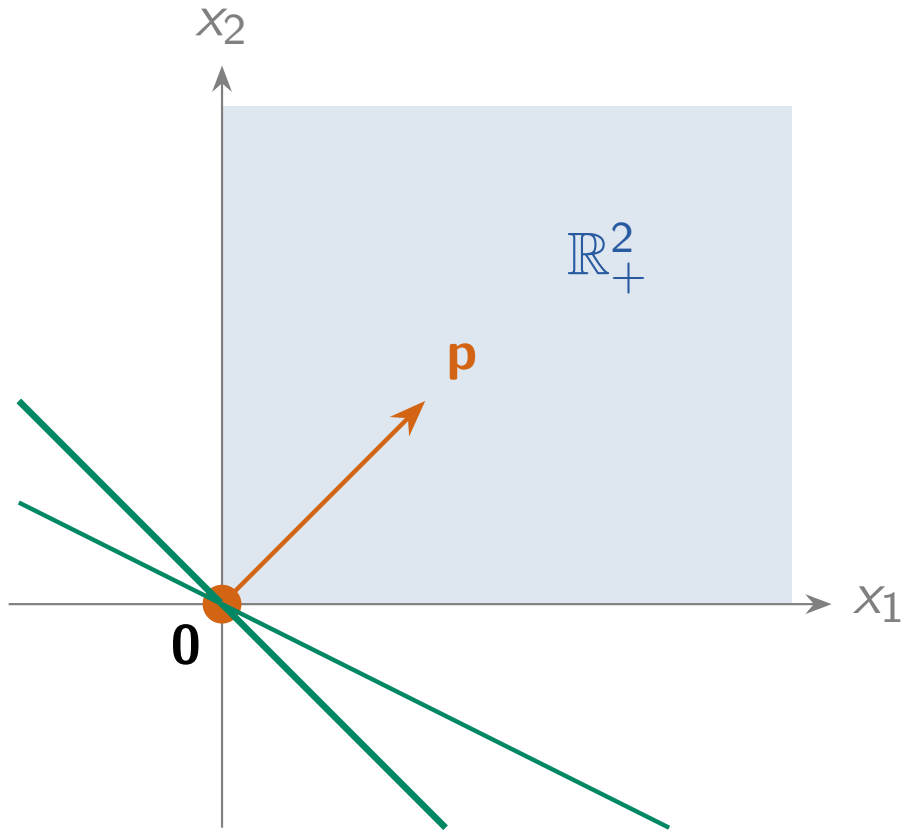
A supporting line must pass through $\mathbf{0}$ and leave the whole quadrant on one side.

Exercise 7.5: Intuition — Supports at the Origin



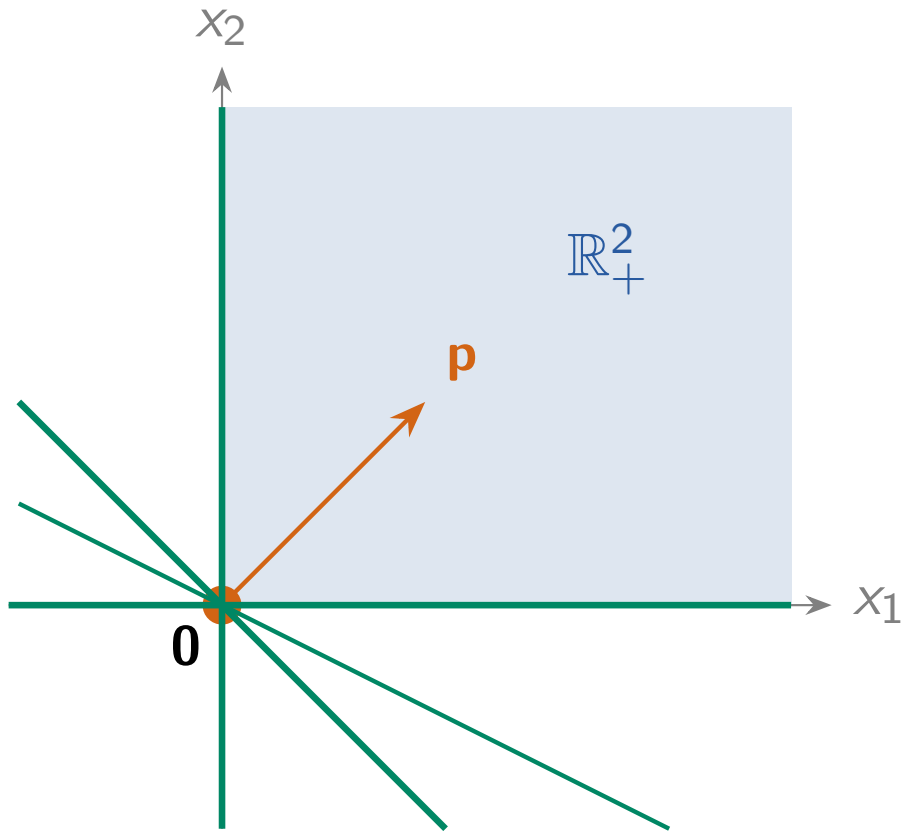
Example: $H((1,1),0)$. The quadrant lies in the halfspace $x_1 + x_2 \geq 0$.

Exercise 7.5: Intuition — Supports at the Origin



Changing the slope gives more supporting lines.

Exercise 7.5: Intuition — Supports at the Origin



The axes also support the quadrant. All supports have a nonzero normal with $p_1, p_2 \geq 0$ or $p_1, p_2 \leq 0$.

Exercise 8.1

Problem

Let $S = \{\mathbf{x}\} \subset \mathbb{R}^3$ be the singleton set of a vector $\mathbf{x} = (1, -1, 0)$. Find the orthogonal complement, which we defined as $S^\perp = \{\mathbf{y} \in \mathbb{R}^3 : \mathbf{y} \cdot \mathbf{x} = 0 \text{ for all } \mathbf{x} \in S\}$. Is $S^\perp$ a linear subspace? What is $(S^\perp)^\perp$?

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Find $S^\perp$ and $(S^\perp)^\perp$. Is $S^\perp$ a subspace of $\mathbb{R}^3$?

Exercise 8.1: Definitions

Find $S^\perp$ and $(S^\perp)^\perp$. Is $S^\perp$ a subspace of $\mathbb{R}^3$?

Definitions

Orthogonal complement: For a set $S \subseteq \mathbb{R}^n$,

$$S^\perp = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{y} \cdot \mathbf{x} = 0 \text{ for every } \mathbf{x} \in S\},$$

where $\mathbf{y} \cdot \mathbf{x} = y_1 x_1 + \cdots + y_n x_n$.

Subspace: A subset S of a linear space X is a subspace if it satisfies
Additivity: $\mathbf{x} + \mathbf{y} \in S$ for every $\mathbf{x}, \mathbf{y} \in S$;
Homogeneity: $\alpha\mathbf{x} \in S$ for every $\mathbf{x} \in S$ and every $\alpha \in \mathbb{R}$.

Exercise 8.1: Sketch of the Proof

Find $S^{\perp}$ and $(S^{\perp})^{\perp}$. Is $S^{\perp}$ a subspace of $\mathbb{R}^3$?

1. Write $\mathbf{y} \cdot \mathbf{x} = 0$ as an equation in y_1, y_2, y_3 and describe $S^{\perp}$.

Exercise 8.1: Sketch of the Proof

Find $S^{\perp}$ and $(S^{\perp})^{\perp}$. Is $S^{\perp}$ a subspace of $\mathbb{R}^3$?

1. Write $\mathbf{y} \cdot \mathbf{x} = 0$ as an equation in y_1, y_2, y_3 and describe $S^{\perp}$.
2. Check additivity and homogeneity.

Exercise 8.1: Sketch of the Proof

Find $S^\perp$ and $(S^\perp)^\perp$. Is $S^\perp$ a subspace of $\mathbb{R}^3$?

1. Write $\mathbf{y} \cdot \mathbf{x} = 0$ as an equation in y_1, y_2, y_3 and describe $S^\perp$.
2. Check additivity and homogeneity.
3. Repeat step 1 with $S^\perp$ in place of S.

Exercise 8.2a

Problem

Consider again the function space $X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$ from problem 3. in PS7.

(a) Define a functional $g : X \to \mathbb{R}$ whose output is the intercept of f, i.e., $g(f) = f(0) = b$. Verify that g is linear on X.

Exercise 8.2a: Definitions

Recall (Exercise 7.3a)

X is a linear space with

$$(f_1 + f_2)(x) = f_1(x) + f_2(x), \qquad (\alpha f)(x) = \alpha f(x).$$

Definitions

Linear functional: A function $g : X \to \mathbb{R}$ is a linear functional if it satisfies

Additivity: $g(f_1 + f_2) = g(f_1) + g(f_2)$ for every $f_1, f_2 \in X$;

Homogeneity: $g(\alpha f) = \alpha g(f)$ for every $f \in X$ and every $\alpha \in \mathbb{R}$.

Exercise 8.2a: Sketch of the Proof

1. Take $f_1(x) = a_1x + b_1$ and $f_2(x) = a_2x + b_2$ in X.

Exercise 8.2a: Sketch of the Proof

1. Take $f_1(x) = a_1x + b_1$ and $f_2(x) = a_2x + b_2$ in X.
2. Check additivity and homogeneity of g.

Exercise 8.2b

Problem

Consider again the function space
$X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$ from problem 3. in PS7.

(b) Show that the set Y, where $b = 1$ from problem 3. in PS7 is a hyperplane in X.

Exercise 8.2b: Definitions

Definitions

Hyperplane: For a nonzero linear functional $f : X \to \mathbb{R}$ and $c \in \mathbb{R}$, the level set

$$H_f(c) = \{x \in X : f(x) = c\}$$

is a hyperplane in X.

Exercise 8.2b: Sketch of the Proof

1. Apply the definition of $H_f(c)$ to the functional g from part (a) to obtain $H_g(c)$.

Exercise 8.2b: Sketch of the Proof

1. Apply the definition of $H_f(c)$ to the functional g from part (a) to obtain $H_g(c)$.
2. Rewrite Y until it has the form $H_g(1)$.

Exercise 8.3

Problem

To show that the convexity of sets S and T is not necessary for hyperplane separation, provide an example of two concrete sets in $\mathbb{R}^2$, none of them convex, and a hyperplane that separates them.

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Find nonconvex $S, T \subseteq \mathbb{R}^2$ and a hyperplane separating them

Exercise 8.3: Definitions

Definitions

Convex set: $S \subseteq \mathbb{R}^n$ is convex if $\alpha \mathbf{x} + (1-\alpha)\mathbf{y} \in S$ for every $\mathbf{x}, \mathbf{y} \in S$ and every $\alpha \in [0,1]$.

Separation: $H(\mathbf{p}, c)$ separates S and T if either

$$\mathbf{p} \cdot \mathbf{x} \leq c \leq \mathbf{p} \cdot \mathbf{y} \quad \text{or} \quad \mathbf{p} \cdot \mathbf{x} \geq c \geq \mathbf{p} \cdot \mathbf{y}$$

for every $\mathbf{x} \in S$ and every $\mathbf{y} \in T$.

Separating hyperplane theorem (Carter, Theorem 3.2)

If S and T are nonempty, disjoint, convex subsets of $\mathbb{R}^n$, then there exists a hyperplane that separates them.

Exercise 8.4

Problem

Minkowski's theorem (Proposition 3.17 in Carter) says that any closed and convex set equals an intersection of closed halfspaces. Let's consider the closed unit ball $B = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$ with the standard norm. Express B as an intersection of halfspaces. To do that, convince yourself that any $\mathbf{p} \in \mathbb{R}^2$ with $\|\mathbf{p}\| = 1$ gives us a supporting hyperplane $H(\mathbf{p}, 1) = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{p} \cdot \mathbf{x} = 1\}$ to B at the point $\mathbf{x} = \mathbf{p}$.

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$$\boxed{\|\mathbf{p}\| = 1 \implies H(\mathbf{p}, 1) \text{ supports } B \text{ at } \mathbf{x} = \mathbf{p}}$$

$$\boxed{B = \bigcap_{\|\mathbf{p}\|=1} H^{-}(\mathbf{p}, 1)}$$

Exercise 8.4: Definitions and Tools

Definitions

Standard norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + x_2^2}$.

Closed halfspaces: $H^+(\mathbf{p}, c) = \{\mathbf{x} : \mathbf{p} \cdot \mathbf{x} \geq c\}$ and $H^-(\mathbf{p}, c) = \{\mathbf{x} : \mathbf{p} \cdot \mathbf{x} \leq c\}$.

Supporting hyperplane: $H(\mathbf{p}, c)$ supports S at $\mathbf{x}_0 \in S$ if $\mathbf{x}_0 \in H(\mathbf{p}, c)$ and $S \subseteq H^+(\mathbf{p}, c)$ or $S \subseteq H^-(\mathbf{p}, c)$.

Triangle inequality

$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Exercise 8.4: Sketch of the Proof

1. For $\|\mathbf{p}\| = 1$, check that $H(\mathbf{p}, 1)$ supports B at $\mathbf{x} = \mathbf{p}$.

▸ Appendix: Cauchy–Schwarz ▸ Appendix: Strict triangle inequality

Exercise 8.4: Sketch of the Proof

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2. Check that $B = \bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1)$.
   - 2.1 Show that $B \subseteq \bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1)$.

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2. Check that $B = \bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1)$.
   - 2.1 Show that $B \subseteq \bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1)$.
   - 2.2 Show that $\bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1) \subseteq B$.

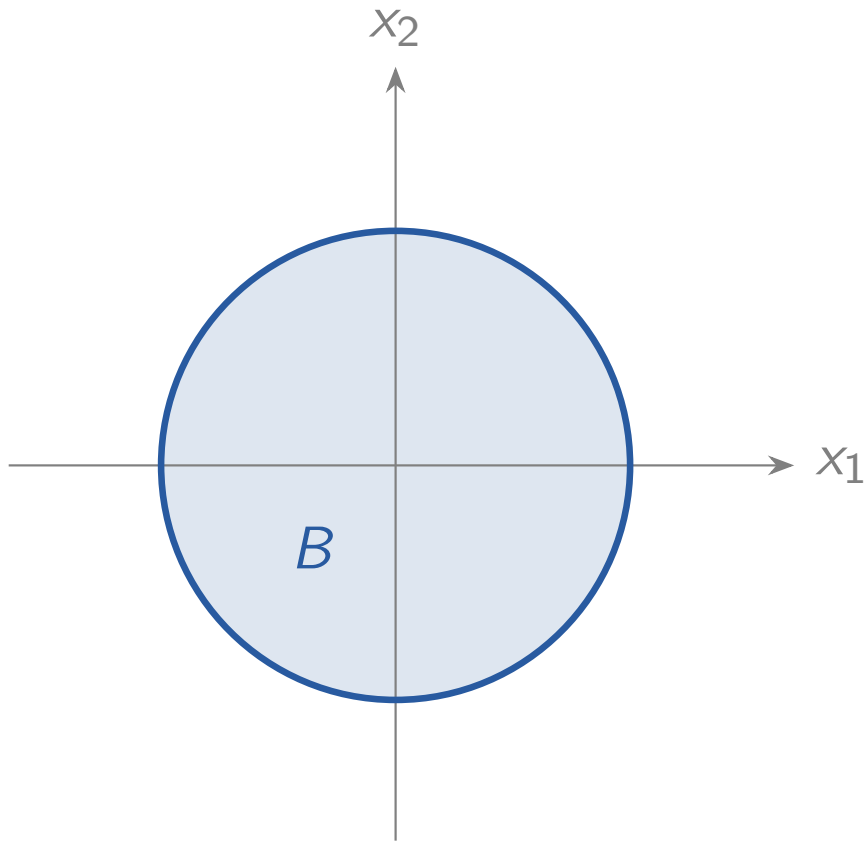
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     - 2.2.1 Use the contrapositive.

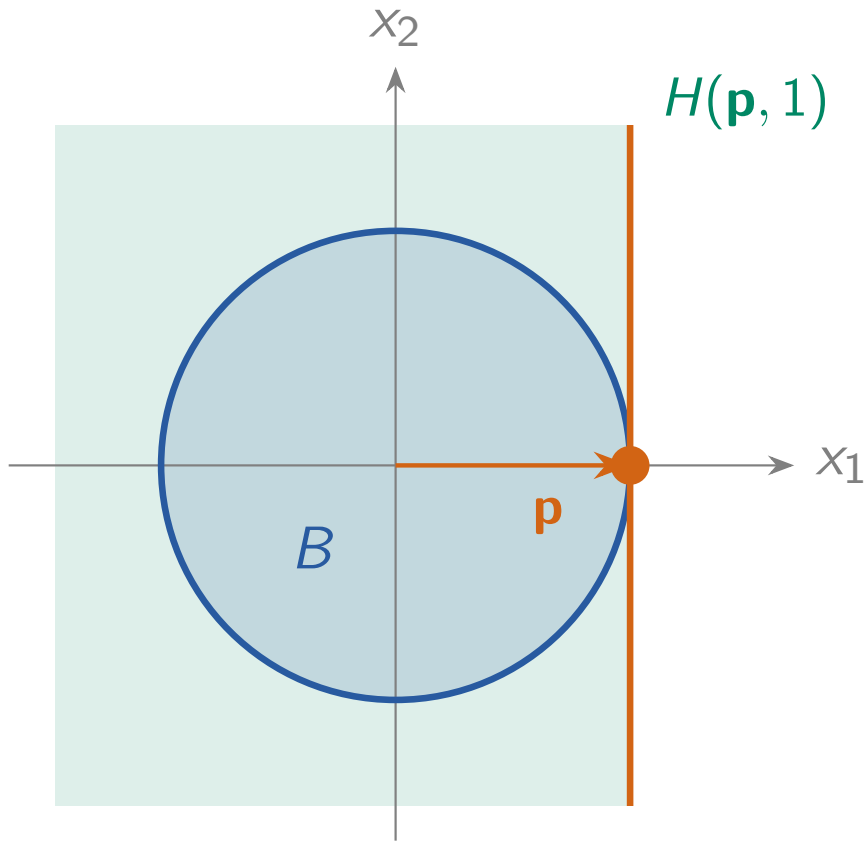
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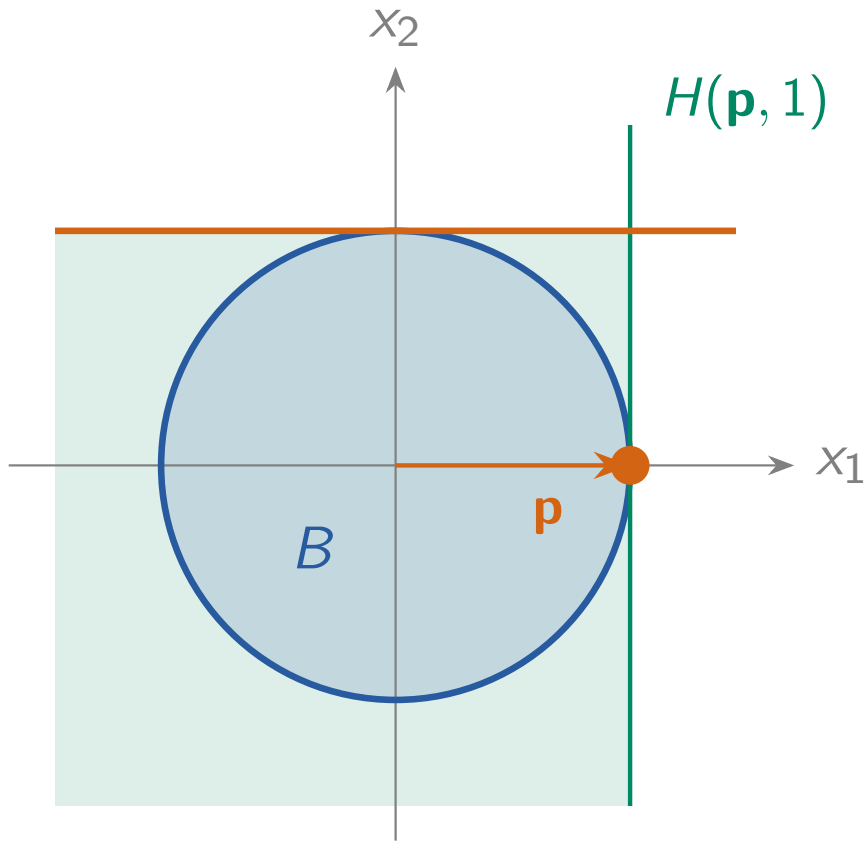
The closed unit ball B in $\mathbb{R}^2$.

Exercise 8.4: Intuition



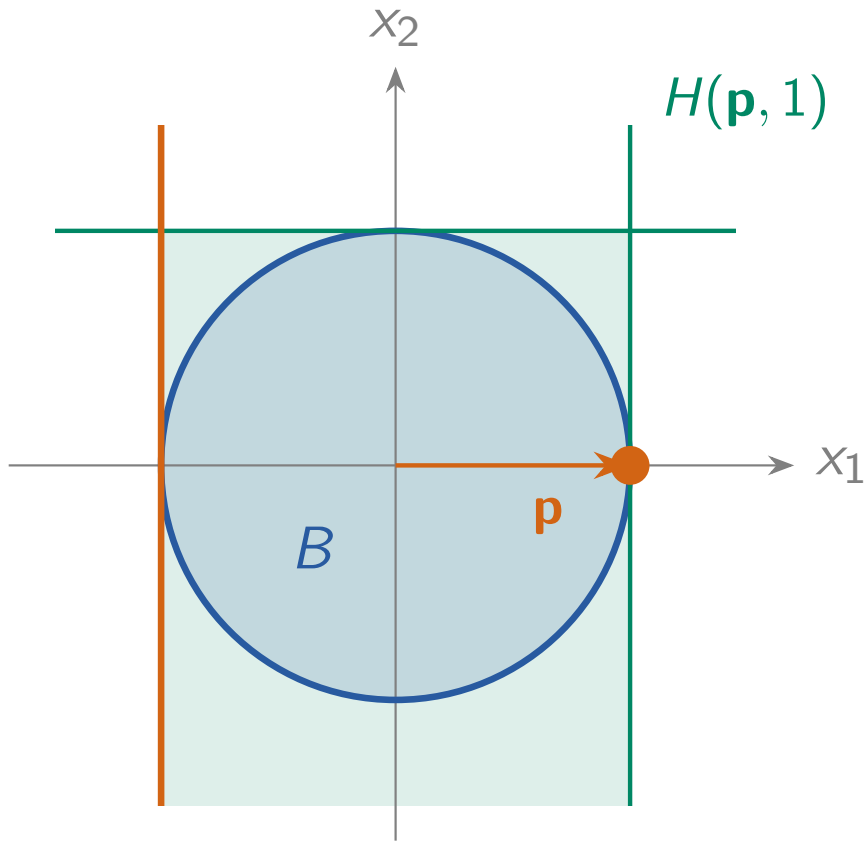
$H(\mathbf{p}, 1)$ touches B at $\mathbf{p}$. Keep the halfspace $H^-(\mathbf{p}, 1)$ containing B.

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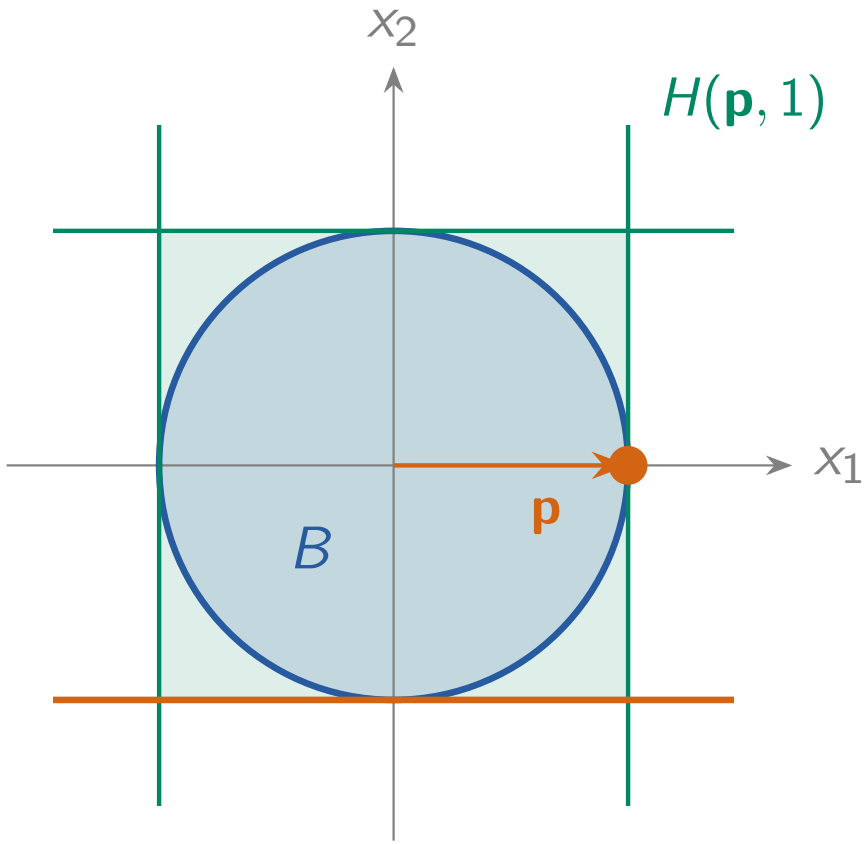
Each new supporting line adds another halfspace. The shaded intersection still contains B.

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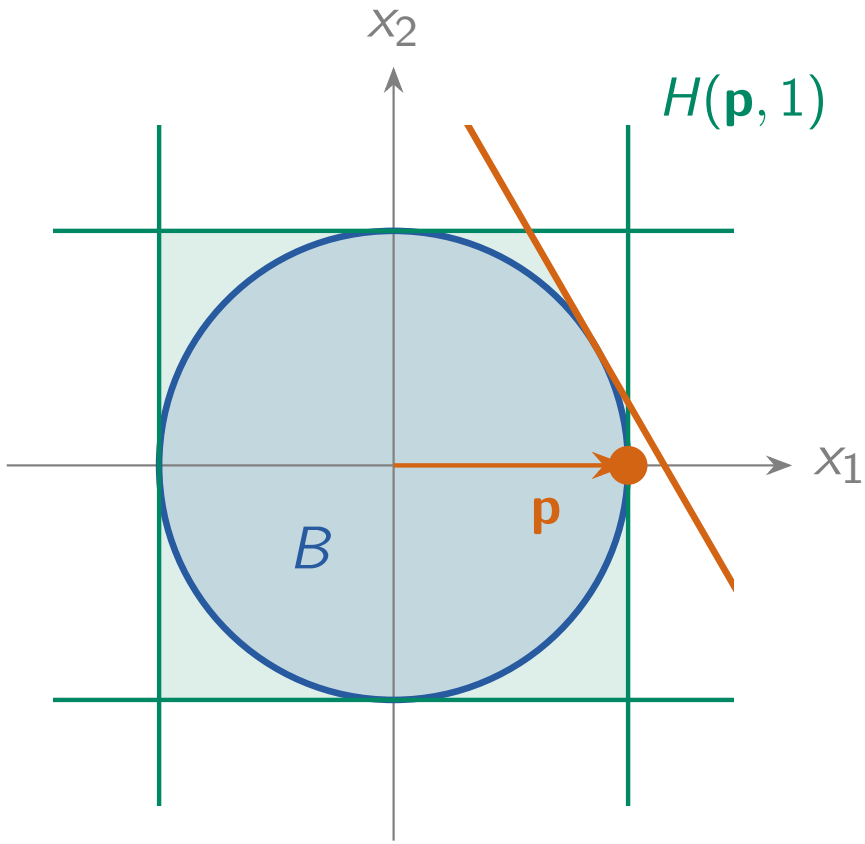
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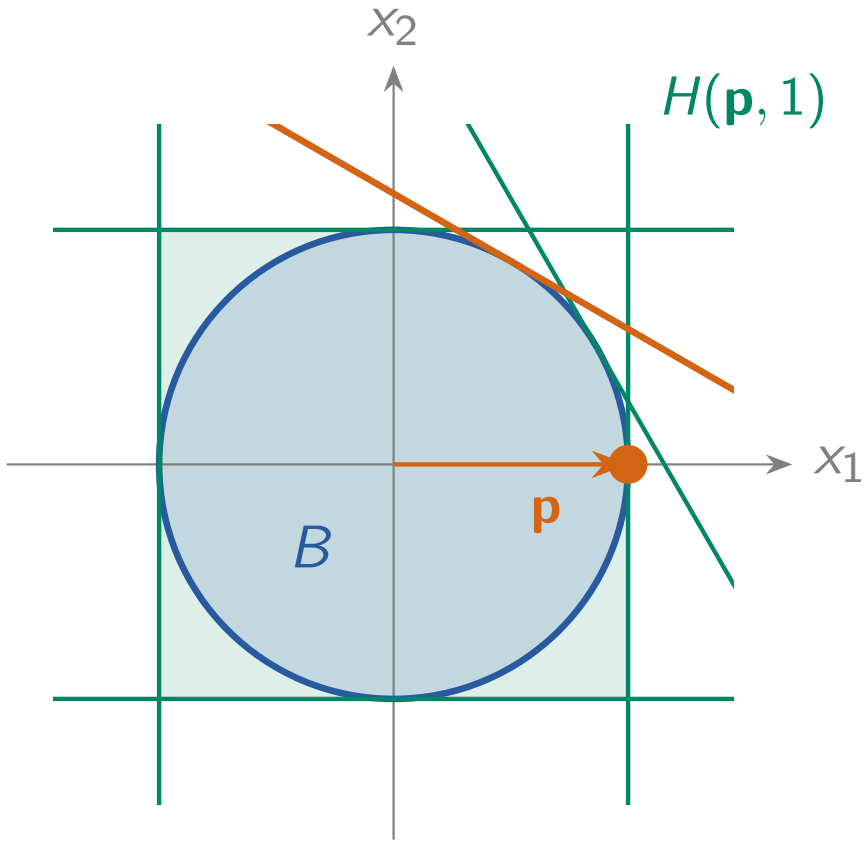
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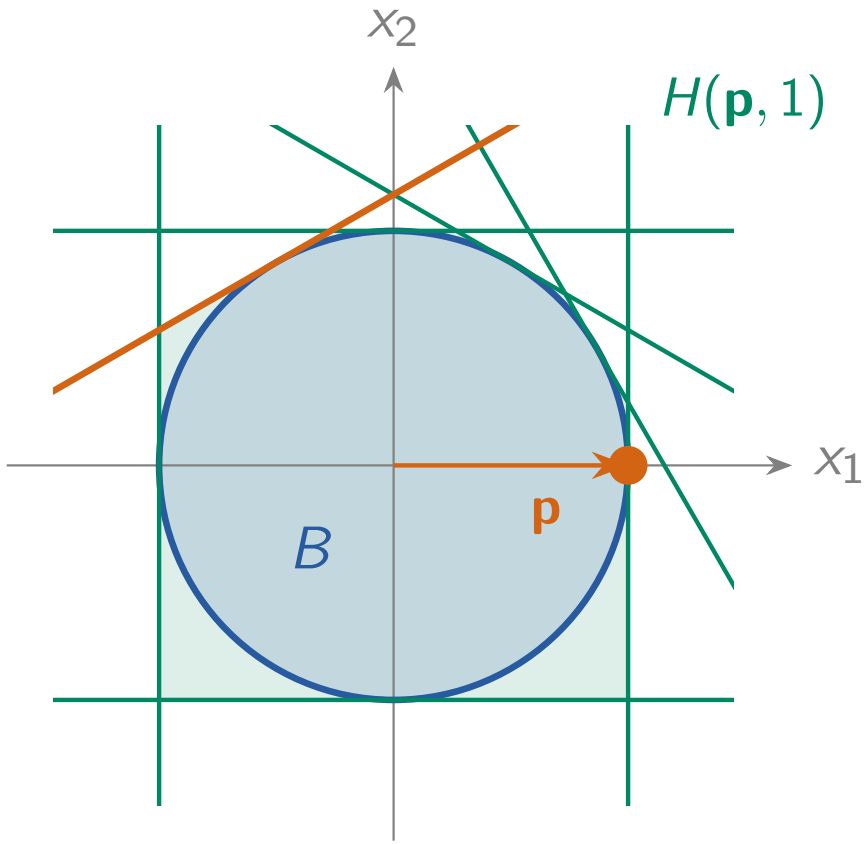
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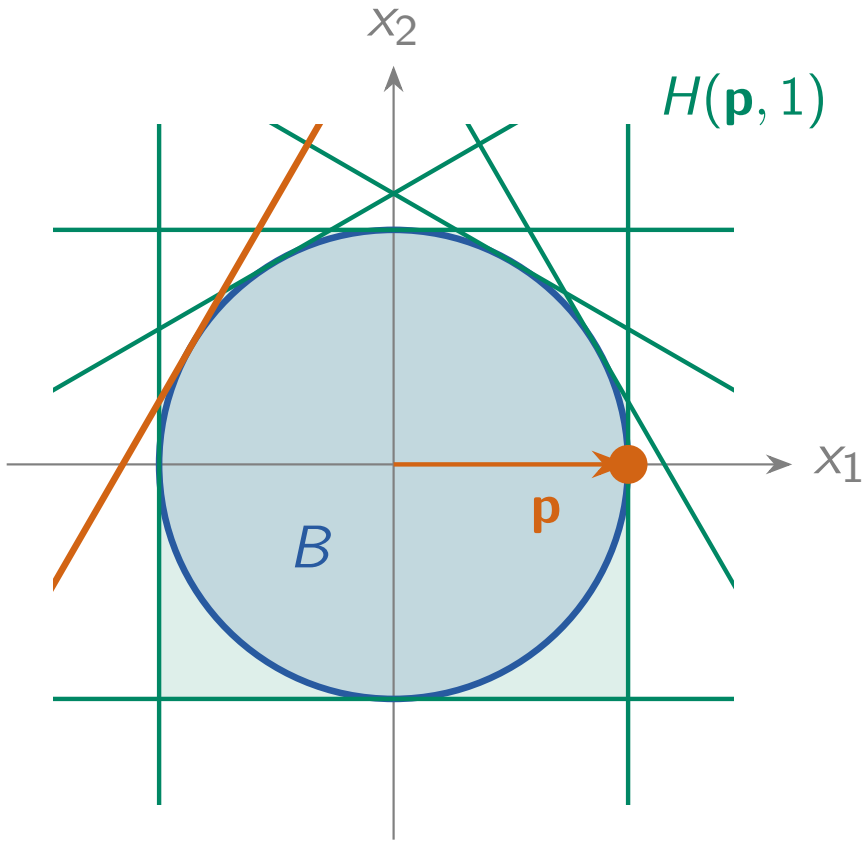
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Exercise 8.4: Intuition



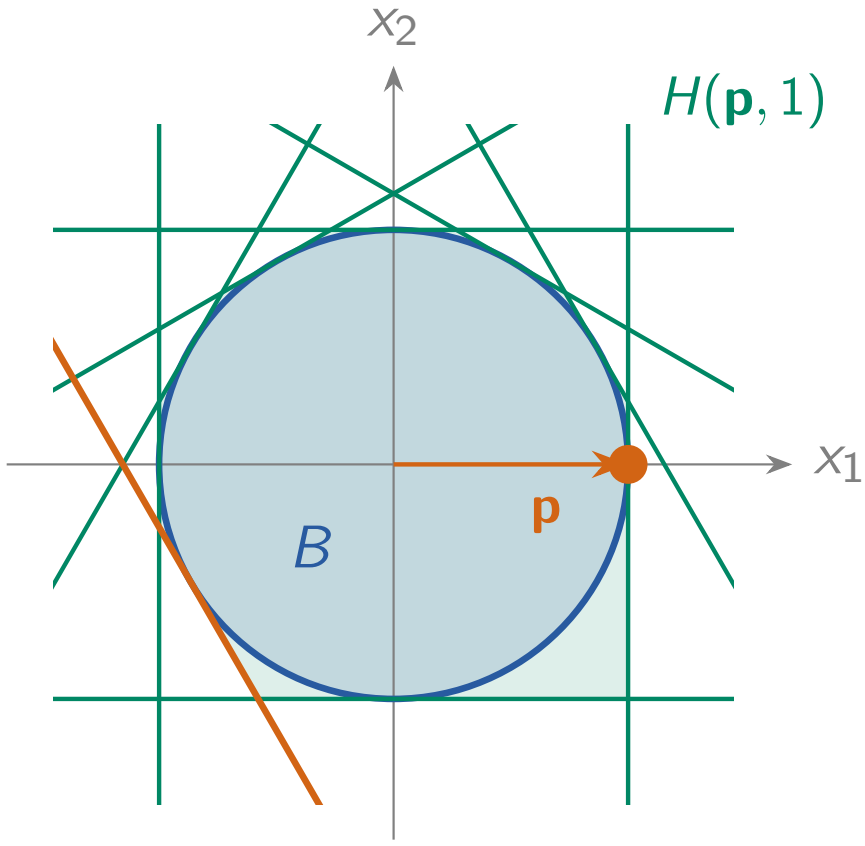
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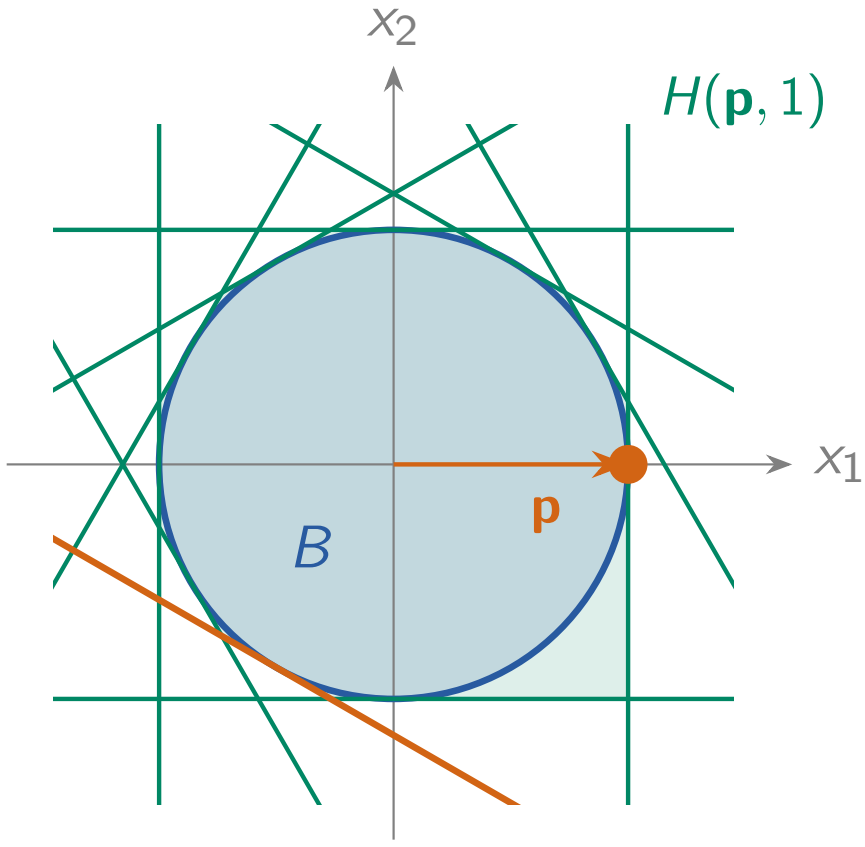
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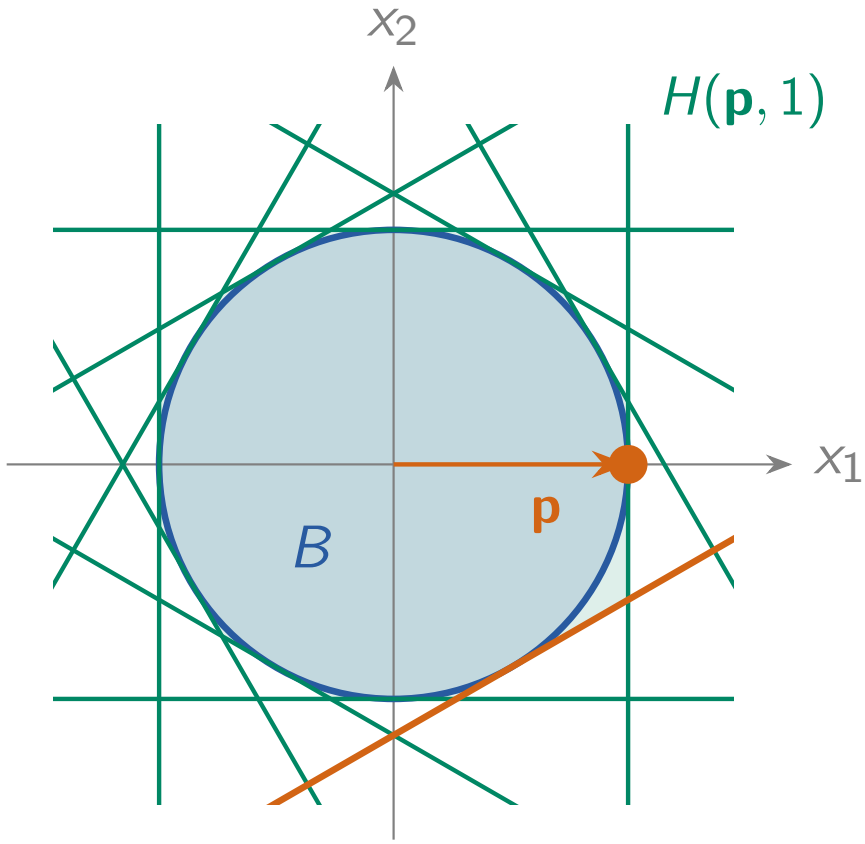
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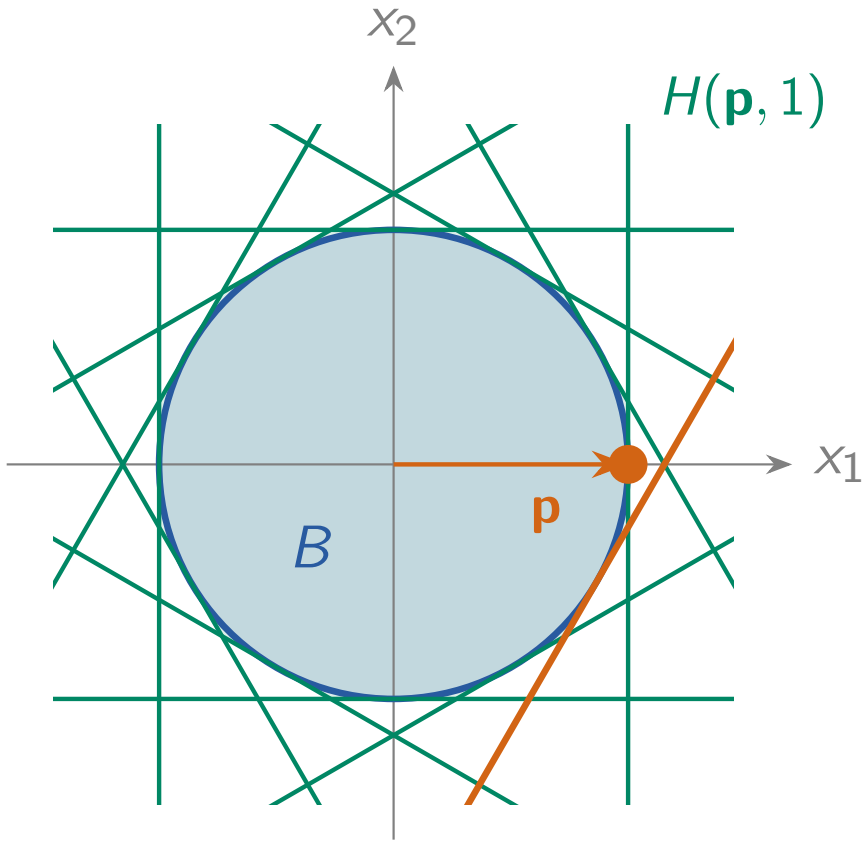
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Exercise 8.4: Intuition



Using every unit normal gives $B = \bigcap_{\|\mathbf{p}\|=1} H^-(\mathbf{p}, 1)$.

Exercise 8.5

Problem

Now let's take $B = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$ as a feasible set in maximization problem. Without setting up a Lagrangian, make use of the supporting hyperplanes in exercise 4. to show that the unique maximizer to the problem

$$\max_{(x_1,x_2)\in B} \quad \alpha x_1 + \beta x_2$$

is $\left(\frac{\alpha}{\|(\alpha,\beta)\|}, \frac{\beta}{\|(\alpha,\beta)\|}\right)$, assuming α or β are not both zero. What would be the solution of the minimization?

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$$\boxed{\mathbf{p} = \frac{(\alpha,\beta)}{\|(\alpha,\beta)\|} \text{ is the unique maximizer of } \alpha x_1 + \beta x_2 \text{ on } B}$$

$$\boxed{\text{Find the minimizer of } \alpha x_1 + \beta x_2 \text{ on } B}$$

Exercise 8.5: Sketch of the Proof — Maximization

$$\max_{(x_1,x_2)\in B} \alpha x_1 + \beta x_2, \qquad (\alpha, \beta) \neq (0,0)$$

1. Use Exercise 8.4 to show that, for $\|\mathbf{p}\| = 1$, $\max_{\mathbf{x}\in B} \mathbf{p}\cdot\mathbf{x} = 1$, with maximizer $\mathbf{x} = \mathbf{p}$.

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4. Rewrite the maximization in step 1 in terms of α and β.
5. Manipulate step 4 to recover the maximization in step 2.

Exercise 8.5: Sketch of the Proof — Uniqueness and Minimum

$$\boxed{\mathbf{x}^* = \mathbf{p} = \frac{(\alpha, \beta)}{\|(\alpha, \beta)\|}}$$

6. Show uniqueness.

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   - 6.2 Show that $\mathbf{y} = \mathbf{x}^*$.

Exercise 8.5: Sketch of the Proof — Uniqueness and Minimum

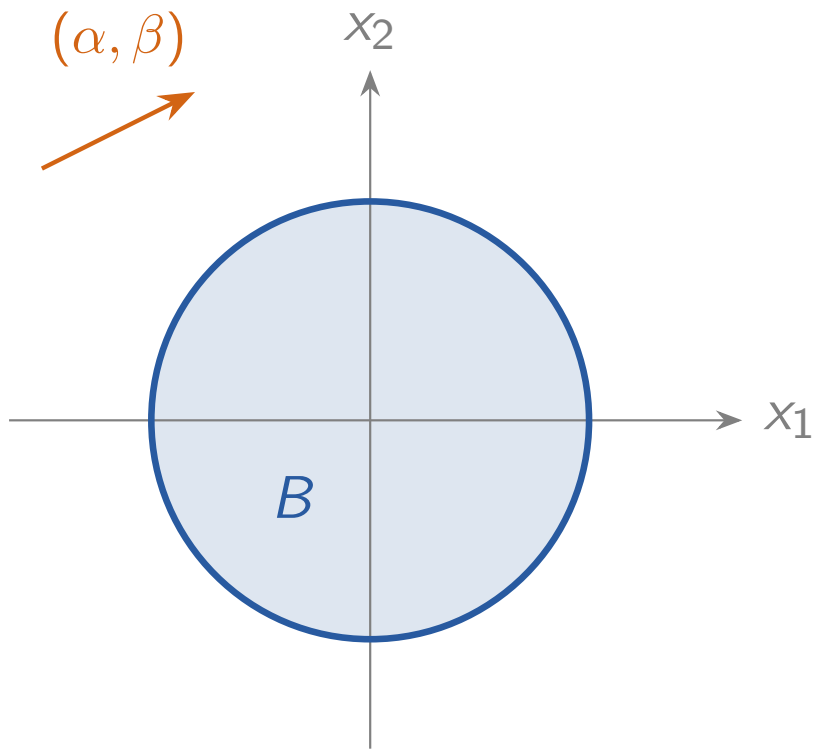
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   - 6.2 Show that $\mathbf{y} = \mathbf{x}^*$.
7. For the minimum, use

$$\min_{\mathbf{x} \in B} (\alpha, \beta) \cdot \mathbf{x} = -\max_{\mathbf{x} \in B} (-\alpha, -\beta) \cdot \mathbf{x}.$$

Exercise 8.5: Intuition

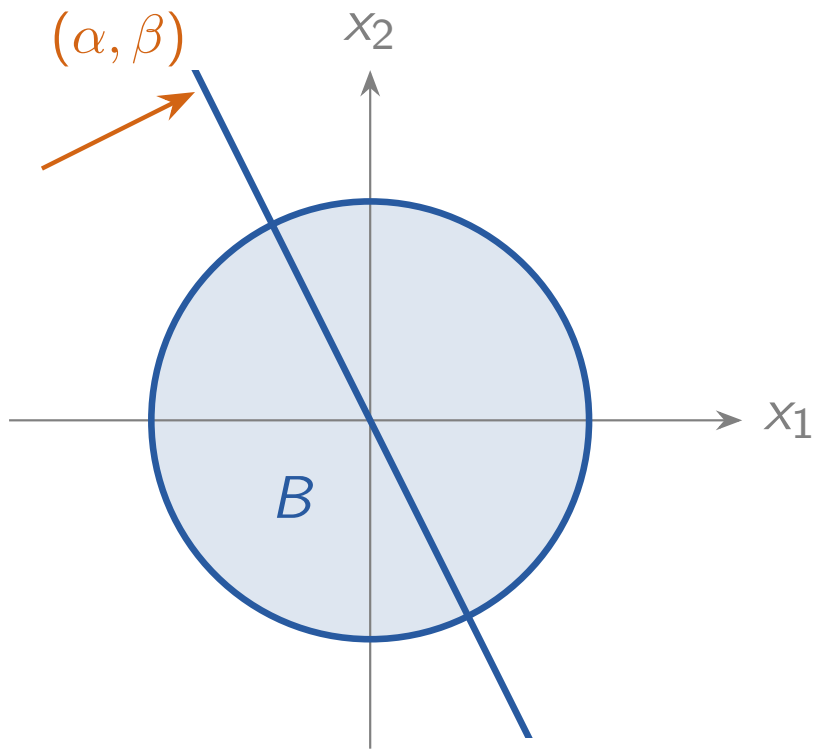
Example: $(\alpha, \beta) = (2, 1)$.



The objective increases in the direction $(2, 1)$.

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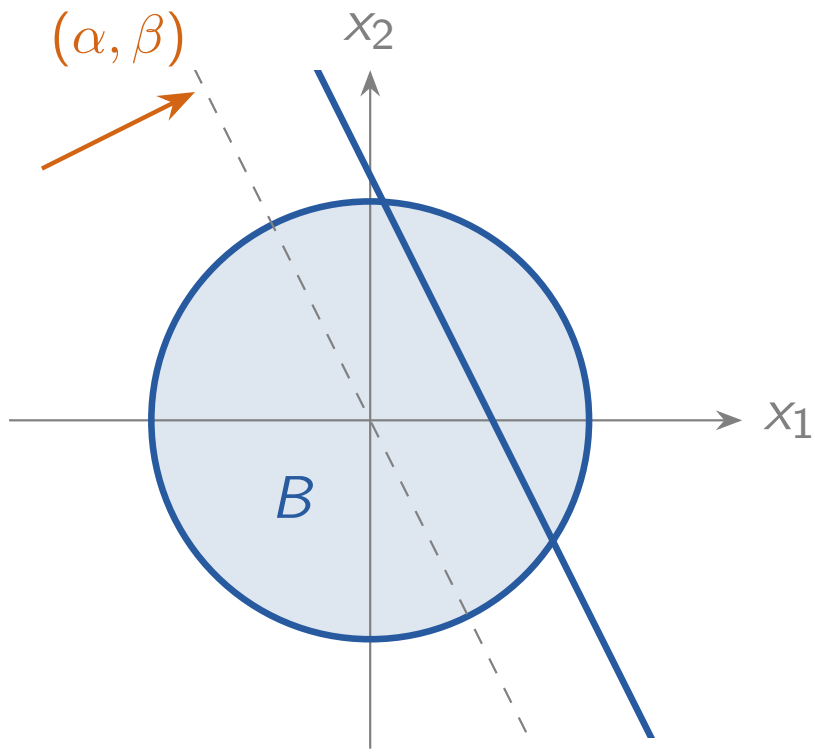
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The level line $2x_1 + x_2 = 0$ crosses B.

Exercise 8.5: Intuition

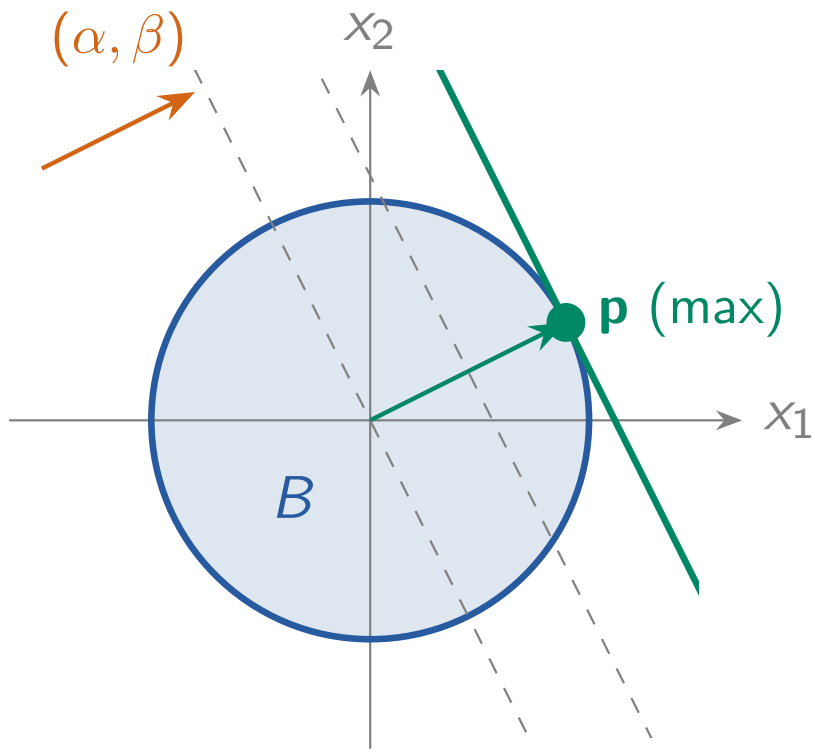
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Increasing the level moves the line in the direction $(2, 1)$.

Exercise 8.5: Intuition

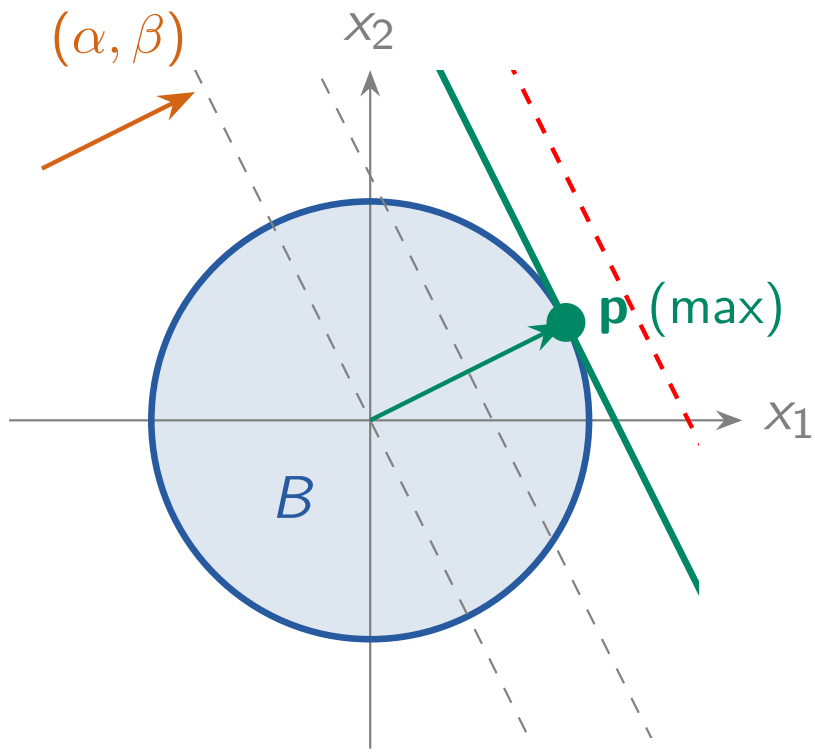
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The maximum is $\sqrt{5}$ at $\mathbf{p} = (2, 1)/\sqrt{5}$.

Exercise 8.5: Intuition

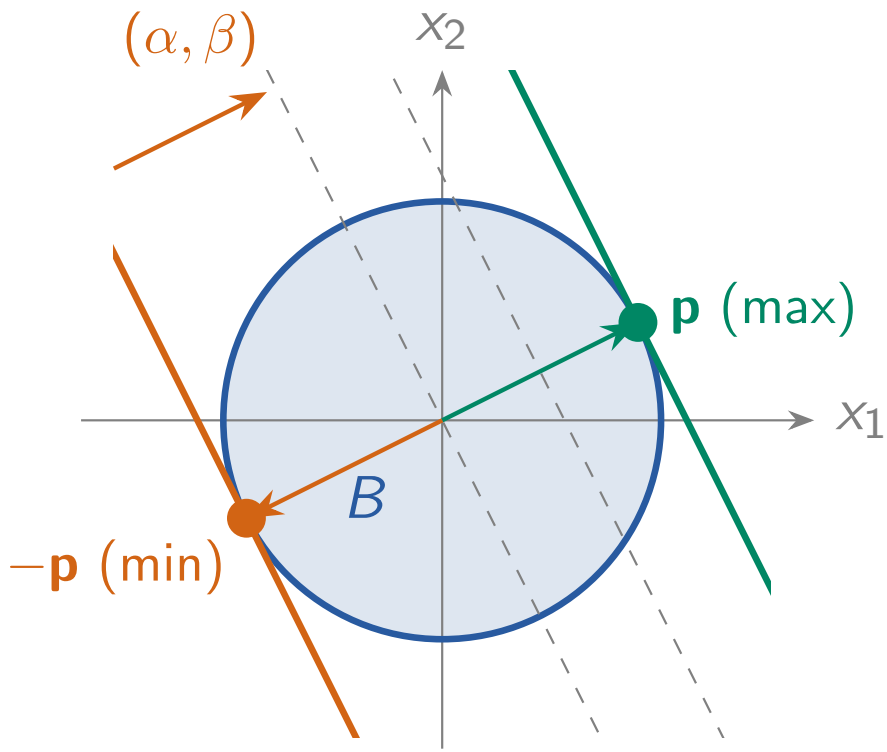
Example: $(\alpha, \beta) = (2, 1)$.



A level above $\sqrt{5}$ has no point in common with B.

Exercise 8.5: Intuition

Example: $(\alpha, \beta) = (2, 1)$.



The minimum is $-\sqrt{5}$ at $-\mathbf{p} = (-2, -1)/\sqrt{5}$.

Appendix: Cauchy–Schwarz Inequality

Cauchy–Schwarz inequality

For every $\mathbf{p}, \mathbf{x} \in \mathbb{R}^n$,

$$|\mathbf{p} \cdot \mathbf{x}| \leq \|\mathbf{p}\| \, \|\mathbf{x}\|.$$

1. Square the triangle inequality:

$$\|\mathbf{p} + \mathbf{x}\|^2 \leq (\|\mathbf{p}\| + \|\mathbf{x}\|)^2.$$

◂ Exercise 8.4: Sketch of the proof

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3. Apply the same argument to $-\mathbf{p}$ and $\mathbf{x}$ to obtain the lower bound.

◂ Exercise 8.4: Sketch of the proof

Appendix: Strict Triangle Inequality

1. Since both sides are nonnegative,

$$\|\mathbf{p}+\mathbf{x}\| < \|\mathbf{p}\| + \|\mathbf{x}\| \quad \Longleftrightarrow \quad \|\mathbf{p}+\mathbf{x}\|^2 < (\|\mathbf{p}\| + \|\mathbf{x}\|)^2.$$

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$$\|\mathbf{p}+\mathbf{x}\|^2 = \|\mathbf{p}\|^2 + 2\mathbf{p}\cdot\mathbf{x} + \|\mathbf{x}\|^2,$$
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4. If $\|\mathbf{p}\| = \|\mathbf{x}\| = 1$, this gives

$$\boxed{\|\mathbf{p}+\mathbf{x}\| < \|\mathbf{p}\| + \|\mathbf{x}\| \quad \Longleftrightarrow \quad \mathbf{p}\cdot\mathbf{x} < 1.}$$

◂ Exercise 8.4: Sketch of the proof